

README for Interim Project

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Overview

This project implements and tests **delta hedging** strategies under the **Black–Scholes model**, using both **simulated data** and **real market data** to evaluate model validity.

The project includes:

- Simulation of stock and option prices under the additive model.
- Calculation of hedging errors and portfolio PNL.
- Validation with **real market data** (GOOG, July 2011).
- Unit tests for implied volatility and delta accuracy.
- Python scripts for visualization.

Files

- Simulation.cpp / Simulation.h – Simulates stock prices, option prices, and hedging errors (Task 1).
- PNL_Cal.cpp / PNL_Cal.h – Real-data delta hedging implementation (Task 2).
- Option_Price.h – Black–Scholes pricing formula and delta calculation.
- utils.h / utils.cpp – Helper functions (date, file handling, etc.).
- unit_test.h – Unit tests for implied volatility and delta.
- plot_simulation.py – Generates Task 1 plots.
- plot_results.py – Generates Task 2 plots.
- result.csv – Real data results (07/05/2011–07/29/2011).
- simulated_stock_prices.csv / simulated_option_prices.csv – Simulation outputs.
- hedging_errors_terminal.csv – Cumulative hedging errors.
- PNG files – Saved plots for reports.

Build & Run

```
g++ -std=c++17 main.cpp -o run.out
```

```
./run.out
```

Generate Plots

```
python3 plot_simulation.py
```

python3 plot_results.py

Unit Test

- Verifies implied volatility search and delta calculation.
- All tests passed successfully.

```
Running Task 2 unit tests...
Running test cases for Task 2...
=====
Option type: C
Searched implied volatility: 0.200000 | Target: 0.200000
Calculated delta: 0.946539 | Target: 0.946539
✅ Test passed!
=====
Option type: P
Searched implied volatility: 0.250000 | Target: 0.250000
Calculated delta: -0.046540 | Target: -0.046540
✅ Test passed!
=====
✅ All test cases passed!

Task 1 simulation...
Generated: simulated_stock_prices.csv, simulated_option_prices.csv,
(.venv) (base) sunyijun@Yijuns-MacBook-Air system_design %
```